Koa Shen

Mechanical Engineering | Mechanical Design, Manufacturing & Testing
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EDUCATION

University of California, Santa Barbara — Expected June 2028
B.S. Mechanical Engineering | 3.97 Major GPA | 3.94 Cumulative GPA | Dean's Honors every quarter — Santa Barbara, CA

- Coursework: Dynamics, Mechanics of Materials, Circuits. Current: Dynamical Systems and Machine Learning.

EXPERIENCE

Mechanical Engineering Intern — Summer 2026 - Present (extended to part-time remote)
SLAC National Accelerator Laboratory - LCLS — Menlo Park, CA

- Built an open-source Python/Drake digital twin for a 30-DOF, 305-body beamline assembly; millisecond collision checks exposed at least five pinch points.
- Developed 15 reusable motion-stage models now being adopted across three LCLS assemblies; applied simulation findings to optics packaging and alignment redesigns.
- Designed mounts and detector strain relief, evaluated stage and mount load cases for $100k+ detectors, and released Solid Edge drawings with GD&T through Teamcenter PDM.

Fabrication Team Lead — Fall 2024 - Present
Gaucho Racing (FSAE EV) - UC Santa Barbara — Santa Barbara, CA

- Led 15 teammates through GR26 driver-safety-system design, fabrication, and installation; the package passed Formula SAE technical inspection.
- CNC-machined 33 suspension components from 4130 steel and 7075-T6 aluminum; held clevis and rod-end bores within a 0.0004-in tolerance window and bearing interfaces within 0.0005 in.
- Worked with designers on GD&T and manufacturability; standardized five spindle interfaces to reuse proven CAM programs and reduce setup and programming work.

Maintenance Technician — Summer 2025
Exploratorium — San Francisco, CA

- Serviced 100+ interactive exhibits and led three major overhauls through failure diagnosis, reverse engineering, manual machining, and welding.
- Redesigned a load-spreading bezel to address acrylic cracking and avoid $3,000+ in replacement costs; documented CAD and remanufacturing instructions for future technicians.

SELECTED PROJECTS

Compact Cycloidal Reducer & Stepper Package — Summer 2026 - Present

- Designed a 15:1 cycloidal reducer within a NEMA 17 motor's 42 mm square footprint; fabricated components and assembled the stepper, driver, and encoder package.
- Built PlatformIO C++ firmware with AS5600 encoder feedback; bench-validated single-joint position correction and missed-step detection.

FSAE Chassis Welding Jig Evolution — Winter 2024 - Spring 2026

- Designed a modular GR25 welding fixture that located 81 chassis tubes within 0.050 in of CAD node positions and reduced assembly time 50% versus the prior fixture.
- Developed a GR26 hybrid fixture with 3D-printed locating brackets and reused 80/20 and plywood, reducing fixture assembly time a further 30%.

FSAE Steering Wheel Development — Fall 2024 - Spring 2026

- Helped develop an FSAE steering wheel integrating buttons, potentiometers, a display, and quick release; iterative, FEA-guided design reduced assembly mass from 5 lb to 1.7 lb.

TECHNICAL SKILLS

Mechanical Design & Analysis: SolidWorks, Solid Edge, Teamcenter PDM, GD&T, DFM, FEA, Statics and dynamics, Kinematics
Manufacturing & Materials: CNC milling and turning, Mastercam, Manual machining, Welding, 4130 steel, 7075-T6 aluminum, 3D printing
Software & Controls: Python, C/C++, MATLAB, Drake, PlatformIO, Stepper motors, AS5600 encoders, Git